Nico Gerardo Bers

bersnico@gmail.com or nicobers2027@u.northwestern.edu | LinkedIn
Winchester, MA & Evanston, IL

Education

- **Northwestern University** *Sept. 2023 - June 2027*
  B.A. in Physics with a concentration in Astronomy; Minor in Machine Learning and Data Science Evanston, IL
  - GPA: 3.9/4.0, 5x on Dean's List
  - Relevant Coursework: Introductory Astrophysics; Classical Hamiltonian and Lagrangian Mechanics; Statistical Mechanics; CS Data Structures and Algorithms; Multivariable Differential and Integral Calculus; Linear Algebra, Fourier Analysis, and Differential Equations.
  - Coursework to be completed by summer 2026: Quantum Mechanics, Advanced Electricity and Magnetism 1, Astrophysical Radiative Processes and Transport (graduate), Intro to Machine Learning, Probability and Statistics for Chemical Engineering.

Research Experience and Employment

- **Northwestern University Dept. of Physics and Astronomy** *Jan. 2026 - Present*
  Undergraduate Research Assistant in Elena Murchikova's group Evanston, IL
  - Searching for gas structures in the galactic center near Sagittarius A* with ALMA radio telescope data.
- **University of Melbourne** *July 2025 - Dec. 2025*
  Science Research Project in Andrew Melatos's group, Member of the LIGO Scientific Collaboration Melbourne, Australia
  - Performing a continuous wave search for the Crab Pulsar in real gravitational-wave data from LIGO's third observing run.
  - Tuning adaptive noise cancellation to improve tracking a neutron star's gravitational-wave frequency with a hidden Markov model's Viterbi decoding algorithm in the presence of non-Gaussian interference.
- **Center for Interdisciplinary Exploration and Research in Astrophysics (CIERA) [🌐]** *Sept. 2023 - July 2025*
  Undergraduate Research Assistant in Vicky Kalogera's group, Member of the LIGO Scientific Collaboration Evanston, IL
  - Gravitational-wave research constraining the redshift distribution of the farthest binary black hole mergers, simulating over 1000 black hole collisions with current LIGO detector sensitivity with hierarchical Bayesian inference and the stochastic background.
  - Developed extended classes of BILBY and GWPopulation libraries for new implementations of the Bayesian templated background search for the stochastic gravitational wave background.
- **Veson Nautical [🌐]** *June 2024 - Sept. 2024*
  Data Science Intern Boston, MA
  - Developed a hierarchical machine learning regression model to estimate the fuel consumption of cargo ships in port stoppages. Improved previous model performance by more than 2 times accuracy.
  - Created a voyage routing estimator for optimizing any maritime voyage using Dijkstra and A* algorithms on weighted graphs with hundreds of thousands of edges and nodes.
- **Veson Nautical [🌐]** *June 2021 - Aug. 2021, June 2022 - Aug. 2022*
  Software Development Intern Boston, MA
  - Developed an iOS app in Swift for nautical fleet tracking and accounting software platform.
  - Responsible for integrating Swift with backend via REST APIs, front-end, and data querying and manipulation with GraphQL and SQL.
- **NASA, TSGC, UT Austin [🌐]** *June 2021 - Aug. 2021*
  STEM Enhancement in Earth Science Research Internship, Earth Systems Group Remote
  - Investigated correlations between mosquito populations and land cover types and contributed to satellite imagery classification.

Papers

- **In preparation** *Jan. 2026*
  Enhancing Searches for the Crab Pulsar in the Second Half of LIGO's Third Observing Run with Adaptive Noise Cancellation
  - N. Bers, A. Melatos et al. *Enhancing Searches for the Crab Pulsar in the Second Half of LIGO's Third Observing Run with Adaptive Noise Cancellation*. In prep. (2026).
- **Published in *The Astrophysical Journal*** *Jan 16, 2026*
  Probing the Peak of Star Formation with the Stochastic Background of Binary Black Hole Mergers
  - N. Bers & A.S. Biscoveanu. *Probing the peak of star formation with the stochastic background of binary black hole mergers*. The Astrophysical Journal (2026) 997: 108. DOI 10.3847/1538-4357/ae2319.

Honors, Scholarships, and Awards

- **2025 USRA Distinguished Undergraduate Honorable Mention** *Oct. 2025*
  Universities Space Research Association [🌐]
  - One of five honorable mentions recognizing contributions to science and US space research, chosen from 130 applicants (5 awardees and 5 honorable mentions).
- **2024-2025 Scholar in Physics & Astronomy** *June 5, 2025*
  Northwestern University Department of Physics and Astronomy [🌐]
  - Received a departmental scholar award for academic and research achievements.
- **NASA Illinois Space Grant Consortium Undergraduate Grant Scholarship Recipient** *Sept. 2024 - June 2025*
  NASA Illinois Space Grant Consortium [🌐]
  - Received a NASA ILSGC scholarship for my academic and research pursuits and successes in STEM.
- **Honorable Mention Rapid Fire Research** *May 31, 2024 & June 6, 2025*
  Northwestern University Department of Physics and Astronomy [🌐][🌐]
  - Received two honorable mentions for my presentation of my research in four minutes or less, competing in a field of graduate and undergraduate researchers.

Leadership, Mentoring, and Teaching Experience

- **Teaching Assistant - Physics 131B: Electricity and Magnetism** *Jan. 2026 - Mar. 2026*
  NU School of Professional Studies/Dept. of Physics and Astronomy Evanston, IL
  - Leading two weekly lab sections and office hours for introductory electricity and magnetism in the School of Professional studies.
  - Explaining foundational physical concepts and lab procedure, and grading weekly lab reports and worksheets.
- **CS 214 Data Structures Peer Mentor** *April 2025 - June 2025*
  Northwestern Computer Science Dept. Evanston, IL[🌐]
  - Held weekly office hours helping students with assignments
  - Responsible for answering lecture questions and grading assignments
- **Club President (2026-27), Executive Board — Interfraternity Council Rep.** *Sept. 2024 - Present*
  Northwestern Survivor Advocacy through Greek Engagement(SAGE) [🌐]
  - Designed new standards aimed at preventing sexual violence within Greek Life.
  - Running speaker events with local organizations advocating for survivors and bringing in donations for the groups.

Oral and Poster Presentations

O=Oral, P=Poster

- **Invited presentation to Monash University Gravity Group [O]** *Oct. 23, 2025*
  Searching for Continuous Gravitational Waves with Adaptive Noise Cancellation: Featuring the Crab Pulsar Melbourne, AUS
- **Final presentation for Science Research Project course[O]** *Oct. 20, 2025*
  Searching for Continuous Gravitational Waves with Adaptive Noise Cancellation: Featuring the Crab Pulsar Melbourne, AUS
- **Rapid Fire Research Physics & Astronomy Dept. Northwestern [O]** *June 6, 2025*
  Exploring the farthest black hole collisions using the gravitational-wave background Evanston, IL
- **34th Midwest Relativity Meeting U. Michigan [O]** *Nov. 17, 2024*
  Probing the peak of star formation with the stochastic background of binary black hole mergers Ann Arbor, MI
- **Rapid Fire Research Physics & Astronomy Dept. Northwestern [O]** *May 31, 2024*
  A Symphony of Binary Black Hole Collisions: Studying Cosmic Noon with Gravitational Wave Backgrounds Evanston, IL
- **Northwestern 2024 Undergraduate Research & Arts Expo. [P]** *May 23, 2024*
  A Symphony of Binary Black Hole Collisions: Studying Cosmic Noon with Gravitational Wave Backgrounds Evanston, IL
- **Chicagoland LIGO-Virgo-KAGRA Meeting [O]** *May 14, 2024*
  Probing the Peak of Star Formation with the Stochastic Background of Binary Black Hole Mergers Evanston, IL
- **American Geophysical Union Fall Meeting, Bright Stars [P]** *Dec. 2021*
  An Analysis of the Effect of Land Cover Type on Mosquito Populations Remote
  - Kamara A., **Bers N.**, et al. (2021, December). **An Analysis of the Effect of Land Cover Type on Mosquito Populations**. In *AGU Fall Meeting Abstracts* (VOL. 2021, PP. ED45B-0637).

Skills

- **Languages:** Bilingual in English and Spanish, professional working Hebrew, limited working French.
- **Programming Languages:** Python, Java, SQL, MATLAB, Mathematica, Swift, DSSL2, Racket/Scheme, Git; some experience with HTML, JavaScript, CSS, React Native.
- **Packages and Libraries:** NumPy, Pandas, Matplotlib, SciPy, BILBY, GWPopulation, NetworkX, Scikit-learn, TensorFlow, XGBoost, CatBoost, LightGBM, Geopandas, Streamlit, and Pydeck. HTCondor, High-Performance Computing (HPC) using the LIGO Data Grid distributed computing centers.
- **Machine Learning:** Regression and classification models, deep neural networks, Convolutional Neural Networks, random forest models, gradient boosting models.